

# Convex Thrackles: a residue bound, a wedge lemma, and the census $2^{n-1} - n$

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## Abstract

A *thrackle* is a geometric graph whose edges pairwise meet, either in a common endpoint or in a proper crossing. Conway conjectured that a thrackle on  $n$  vertices has at most  $n$  edges; for vertices in convex position (a *convex thrackle*) this is classical and admits a one-paragraph proof. We give that proof in a form we could not find in the literature: encoding each chord  $\{v_i, v_j\}$  by its *residue*  $i + j \bmod n$  gives an injection from the edges of any convex thrackle into  $\mathbb{Z}/n\mathbb{Z}$  (Lemma 2.1), which also characterises the extremal case. Using the same encoding we obtain a complete description of the interaction between a convex thrackle and the boundary of the polygon (Theorem 3.1, 3.2), which in particular shows that the  $n$  *doublestars* are the only convex thrackles containing two polygon edges.

Our main contribution is a census. We prove (Theorem 4.1) that a set of  $m$  convex points supports a *unique* Hamiltonian thrackle cycle when  $m$  is odd — the cycle in which every vertex is joined to the vertex almost opposite to it — and (Theorem 4.3) that every off-cycle vertex has a forced, unique way of attaching. Consequently maximal convex thrackles are in bijection with the odd-sized subsets of size at least 3 of the vertex set, and we obtain the closed form

$$\#\{\text{maximal convex thrackles on } n \text{ convex points}\} = 2^{n-1} - n.$$

The count is refined: exactly  $\binom{n}{m}$  of them have cycle of length  $m$ .

We did not find a prior statement of this count in the literature; §2 and §8 document what is known (in particular that the closely related cover number  $\chi(D_n)$  is *not* new and is known in closed form), and state precisely which parts of our argument are novel and which are standard. All results are verified exhaustively by machine for  $n \leq 11$ , by two independent routes, and we report the code.

**Keywords:** combinatorial geometry; thrackles; convex position; extremal combinatorics; enumeration; exact cover.

## 1 Introduction

Fix  $n \geq 3$  labelled points  $v_0, v_1, \dots, v_{n-1}$  in convex position in the plane, in this cyclic order (indices modulo  $n$ ). A *chord* is an unordered pair  $\{v_i, v_j\}$ ; there are  $\binom{n}{2}$  of them, of which  $n$  are *polygon edges* and  $n(n-3)/2$  are diagonals. Two chords *meet* if they share a vertex or cross in the interior of the polygon, and a set of chords in which every two members meet is a *convex thrackle*. Such a set is written  $\mathcal{T}$  and we write  $E(\mathcal{T})$  for its edges; throughout the paper “thrackle” means convex thrackle and “edge” means chord.

Two parameters have attracted attention: the maximum number of edges of a thrackle (Conway’s problem) and the minimum number of thrackles needed to cover all chords of the polygon (the chromatic number of the convex segment disjointness graph). Both are understood. Conway’s conjecture is open in general (best current bound  $1.393n$  due to Xu [11]); for convex

position the bound  $|E| \leq n$  is classical and easy. The cover number has the exact value

$$\chi(D_n) = n - \left\lfloor \sqrt{2n + \frac{1}{4}} - \frac{1}{2} \right\rfloor,$$

proved by Araujo, Dumitrescu, Hurtado, Noy and Urrutia [1] for lower bounds and by Fabila-Monroy and Wood [3] and Dujmovic and Wood [2] for the matching construction.

This paper sits between those two results. It is a study of the *individual* convex thrackles themselves: what they can look like, how many there are, and how they interact with the boundary of the polygon.

## 1.1 Contributions

1. **A residue encoding** (Lemma 2.1). The map  $e = \{v_i, v_j\} \mapsto i + j \bmod n$  is injective on the edges of every convex thrackle. This gives a two-line proof of  $|E(\mathcal{T})| \leq n$  and a criterion for equality. We are not aware of this formulation in the thrackle literature.
2. **Boundary structure** (Theorem 3.1, 3.2). A convex thrackle on  $n \geq 4$  points contains at most two polygon edges; if it contains two they are consecutive and  $\mathcal{T}$  is contained in the *doublestar* at their common vertex (the star at that vertex plus the chord joining its two neighbours), which has exactly  $n$  edges; there are exactly  $n$  such thrackles. A convex thrackle containing exactly one polygon edge has at most  $n$  edges, with a complete characterisation of equality.
3. **The census** (Theorem 4.1–4.4). Each odd set  $S$  of size  $m \geq 3$  supports a unique Hamiltonian thrackle cycle; each vertex outside  $S$  has a forced and unique way of attaching to it; hence maximal convex thrackles are in bijection with the odd subsets of size  $\geq 3$ , and their number is exactly  $2^{n-1} - n$ , refined to  $\binom{n}{m}$  of cycle length  $m$ .
4. **Size-restricted cover numbers** (Section 6). We introduce  $\chi_s(n)$ , the minimum number of thrackles of size at most  $s$  covering all chords, prove two elementary bounds, and compute it exactly for  $n \leq 10$ .
5. **Machine verification** (Section 7). Every theorem is verified exhaustively for  $n \leq 11$  by two independent routes (brute-force clique enumeration and a constructive enumeration from odd subsets), together with a dedicated falsification suite (§9). As an external check on the pipeline we also recompute  $\chi(D_n)$  by integer programming and reproduce the published closed form for all  $n \leq 11$ .

## 1.2 What is not claimed

We do not claim novelty for  $|E(\mathcal{T})| \leq n$ , for the pseudoforest structure of thrackles, or for  $\chi(D_n)$ : these are known (§2). In particular an earlier version of this work conjectured that  $\chi(D_n) = \lfloor 2n/3 \rfloor$ ; the literature check in §2 refuted this, the conjecture was wrong for large  $n$  (since  $\chi(D_n) \sim n - \sqrt{2n}$ ), and we record the episode in §9.3 as a worked example of a falsification that mattered. Lemma 2.1 and Theorem 4.4 are our own claims; for those we say only that we did not find a prior statement.

# 2 Related work and known results

## 2.1 Thrackles

Conway [6] conjectured that every thrackle on  $n$  vertices has at most  $n$  edges. Erdős [7] proved this for thrackles drawn with straight segments, and Perles gave another proof; Woodall [12] extended it to  $x$ -monotone and outerplanar drawings, and Cairns and Nikolayevsky [?, 5]

formalised the structure of outerplanar (equivalently convex) thrackles. For general thrackles the best upper bound is  $1.393n$  [11], improving  $167n/117$  of Fulek and Pach [8]. Each connected component of a linear thrackle contains at most one cycle and it is odd, which is the usual route to the bound  $|E| \leq |V|$  [10].

For convex thrackles specifically, the structural description is standard: a maximal convex thrackle is an odd cycle together with degree-one vertices attached to it, and each off-cycle vertex lies in a unique “wedge” determined by the cycle [1, 2]. This is the structure we make precise and count in §4.

## 2.2 Cover numbers of chords

Let  $D_n$  be the graph whose vertices are the chords of a convex  $n$ -gon, with two chords adjacent when they are *disjoint*. A colour class of  $D_n$  is exactly a convex thrackle, so  $\chi(D_n)$  is the minimum number of convex thrackles covering all chords. Araujo, Dumitrescu, Hurtado, Noy and Urrutia [1] proved  $\chi(D_n) \geq n - \sqrt{2n}$  together with the sharp bound  $kn - \binom{k}{2}$  on the union of  $k$  maximal convex thrackles; Dujmovic and Wood [2] and Fabila-Monroy and Wood [3] established the matching construction, giving the closed form cited above. We reproduce this value computationally in §7.3 as a validation of our own pipeline, and we do not treat it as a contribution.

## 2.3 Enumeration of geometric structures in convex position

Counting non-crossing graphs, thrackles and related structures on convex point sets is a well-developed area [1]. Flajolet and Noy [9] give an analytic theory for non-crossing configurations. We are not aware of a published enumeration of convex thrackles by size; §4.4 and §7.1 give one.

## 3 Preliminaries

**Definition 3.1** (convex thrackle). A *convex thrackle* on  $\mathcal{V} = \{v_0, \dots, v_{n-1}\}$  is a set  $\mathcal{T}$  of chords of the convex  $n$ -gon such that every two members of  $\mathcal{T}$  meet. It is *maximal* if no chord can be added while keeping this property.

Two elementary facts about chords of a convex polygon are used constantly. Write  $e = \{v_i, v_j\}$  with  $i < j$  and  $f = \{v_a, v_b\}$  with  $a < b$ . If  $e$  and  $f$  are vertex-disjoint then they cross iff their endpoints alternate around the polygon, i.e. iff  $i < a < j < b$  or  $a < i < b < j$ . We call the open arcs cut by the chord  $\{v_p, v_q\}$  the two sets of points strictly between  $v_p$  and  $v_q$  in the two directions around the polygon.

**Lemma 3.2** (crossing criterion). *Let  $e = \{v_p, v_q\}$  and let  $f = \{v_a, v_b\}$  be vertex-disjoint. Then  $e$  and  $f$  cross iff exactly one of  $v_a, v_b$  lies in each of the two open arcs cut by  $e$ .*

**Definition 3.3** (residue). For a chord  $e = \{v_i, v_j\}$  of a convex  $n$ -gon,  $\rho_n(e) = i + j \in \mathbb{Z}/n\mathbb{Z}$  is its *residue*.

## 4 The residue bound

**Lemma 4.1** (injectivity of the residue map). *Let  $\mathcal{T}$  be a convex thrackle on  $\mathcal{V}$ . Then  $\rho_n$  is injective on  $E(\mathcal{T})$ . In particular  $|E(\mathcal{T})| \leq n$ , with equality iff every residue class occurs exactly once among the edges of  $\mathcal{T}$ .*

*Proof.* Suppose  $e = \{v_i, v_j\} \neq f = \{v_a, v_b\}$  are edges of  $\mathcal{T}$ , with  $i < j$  and  $a < b$ , and  $i + j \equiv a + b \pmod{n}$ . We first note that  $e$  and  $f$  cannot share a vertex. If  $j = a$  then  $i + j \equiv a + b = j + b \pmod{n}$  forces  $i \equiv b$ ; since  $0 \leq b - j < i$  and  $i < b$  is impossible ( $a = j < b$  and  $j > b$  cannot both hold, because  $j = a < b$  gives  $j < b$ ), the congruence gives  $b = i$ , contradicting  $i < j < b$ . The case  $i = b$  is symmetric, and  $j = b$  or  $i = a$  would give  $e = f$ . So  $e$  and  $f$  are vertex-disjoint and, as members of a thrackle, they cross. Interchanging  $e, f$  if necessary assume  $i < a$ . Crossing forces  $i < a < j < b$  (the alternative  $a < i < b < j$  contradicts  $i < a$ ).

Hence  $a + b - (i + j) = (a - i) + (b - j) \geq 2 > 0$ , and since  $i + j \equiv a + b \pmod{n}$  with  $0 < a + b - i - j \leq (n - 2) + (n - 1) - (i + j) < 2n$  we get

$$a + b = i + j + n. \quad (1)$$

But  $b \geq j + 1$  and  $a \leq j - 1$  (as  $a < j$ ), so  $a + b \leq j - 1 + n - 1 = j + n - 2$ . Together with (1) this gives  $i + j + n \leq j + n - 2$ , i.e.  $i \leq -2$ , a contradiction.  $\square$

**Corollary 4.2.** *Every convex thrackle on  $n$  convex points has at most  $n$  edges, and it has exactly  $n$  edges iff its residues exhaust  $\mathbb{Z}/n\mathbb{Z}$ .*

**Remark 4.3.** The bound itself is not new: it is the convex case of Conway's conjecture, known since Erdős [7]. What we contribute here is the residue encoding, which gives a two-paragraph proof with no appeal to geometry beyond Lemma 3.2, and which also pins down the extremal case in a usable form (Corollary ??). We did not find this formulation in the literature.

**Lemma 4.4** (doublestars attain the bound). *For each  $v \in \mathcal{V}$  the doublestar*

$$D_v := \{\{v, w\} : w \in \mathcal{V} \setminus \{v\}\} \cup \{\{v_{v-1}, v_{v+1}\}\}$$

*(indices mod  $n$ ) is a convex thrackle with exactly  $n$  edges, and the  $n$  doublestars are pairwise distinct.*

*Proof.* All edges but one meet at  $v$ , and they clearly meet each other. The remaining edge  $\{v_{v-1}, v_{v+1}\}$  meets every star edge: it meets  $\{v, v_{v-1}\}$  and  $\{v, v_{v+1}\}$  at an endpoint, and for every other  $w$  the chord  $\{v_{v-1}, v_{v+1}\}$  crosses  $\{v, w\}$  because exactly one of  $v_{v-1}, v_{v+1}$  lies in each open arc cut by it. So  $|E(D_v)| = n$ . Distinctness:  $D_v$  contains the  $n - 1$  star edges at  $v$  and only the single non-star edge  $\{v_{v-1}, v_{v+1}\}$ , so the set of vertices of degree  $\geq 2$  determines  $v$  (for  $n \geq 4$ ).  $\square$

## 5 Interaction with the boundary of the polygon

**Theorem 5.1** (at most two polygon edges). *Let  $n \geq 4$  and let  $\mathcal{T}$  be a convex thrackle on  $\mathcal{V}$ . Then  $\mathcal{T}$  contains at most two polygon edges. If it contains exactly two, they are  $\{v_{v-1}, v_v\}$  and  $\{v_v, v_{v+1}\}$  for some  $v$ , and  $\mathcal{T} \subseteq D_v$ . Consequently the convex thrackles containing two polygon edges are exactly the  $n$  maximal ones  $D_v$ .*

*Proof.* Polygon edges never cross anything, so two polygon edges meet iff they are consecutive, say  $p = \{v_{v-1}, v_v\}$  and  $q = \{v_v, v_{v+1}\}$ ; three polygon edges can never be pairwise consecutive. Suppose  $\mathcal{T}$  contains  $p$  and  $q$ . Any other edge  $e \in \mathcal{T}$  must meet both  $p$  and  $q$ . Meeting a polygon edge is possible only by sharing an endpoint with it (nothing crosses a boundary edge), so  $e$  has an endpoint in  $\{v_{v-1}, v_v\}$  and an endpoint in  $\{v_v, v_{v+1}\}$ . Since  $v_v$  is itself a possible endpoint, the possibilities for  $e$  are:  $e = p$ ,  $e = q$ ,  $e = \{v_{v-1}, v_{v+1}\}$ , or  $e$  is incident with  $v_v$ . So every edge of  $\mathcal{T}$  lies in  $D_v$ , and since  $D_v$  is a thrackle (Lemma 4.4) and  $\mathcal{T}$  is maximal iff  $|E(\mathcal{T})| = n$ , the maximal ones are exactly the  $D_v$ .  $\square$

**Theorem 5.2** (exactly one polygon edge). *Let  $n \geq 4$ ,  $1 \leq i \leq n-1$  and put  $w_r = v_{(i+1+r) \bmod n}$  for  $1 \leq r \leq n-2$ . For  $1 \leq t \leq n-2$  let*

$$S(i, t) := \{v_i v_{i+1}\} \cup \{v_i w_t, v_{i+1} w_t\} \cup \{v_i w_r : 1 \leq r < t\} \cup \{v_{i+1} w_s : t < s \leq n-2\}.$$

*Then  $S(i, t)$  is a convex thrackle with exactly  $n$  edges whose only polygon edge is  $\{v_i v_{i+1}\}$ . Conversely, every convex thrackle containing exactly one polygon edge  $\{v_i v_{i+1}\}$  is contained in a unique  $S(i, t)$  and has at most  $n$  edges, with equality only if it equals some  $S(i, t)$ .*

*Proof.* Fix  $p = v_i$ ,  $q = v_{i+1}$  and let  $A$  be the set of  $j \in \{1, \dots, n-2\}$  with  $\{pw_j\} \in \mathcal{T}$ ,  $B$  the set of  $s$  with  $\{qw_s\} \in \mathcal{T}$ . Every edge of  $\mathcal{T} \setminus \{pq\}$  meets  $pq$ , hence has an endpoint in  $\{p\}$  or in  $\{q\}$ , so  $\mathcal{T} = \{pq\} \cup \{\{p, w_j\} : j \in A\} \cup \{\{q, w_s\} : s \in B\}$ . For  $j \in A$  and  $s \in B$  with  $j \neq s$  the chords  $\{pw_j\}$  and  $\{qw_s\}$  are vertex-disjoint and, using Lemma 3.2 with the order  $p, q, w_1, w_2, \dots, w_{n-2}$ , they cross iff  $j < s$ . Hence every element of  $A \setminus B$  is smaller than every element of  $B \setminus A$ . If  $|A \cap B| \geq 2$ , say  $j < s$  both in  $A \cap B$ , then  $\{pw_j\}$  and  $\{qw_s\}$  would have to cross, which by the above requires  $j < s$ , while  $\{pw_s\}$  and  $\{qw_j\}$  would have to cross too, requiring  $s < j$  — a contradiction. So  $|A \cap B| \leq 1$ .

*Case  $|A \cap B| = 1$ , say  $A \cap B = \{w_t\}$ .* Then  $A \setminus \{t\} \subseteq \{1, \dots, t-1\}$  and  $B \setminus \{t\} \subseteq \{t+1, \dots, n-2\}$ , so  $|A| + |B| \leq (t-1) + (n-2-t) + 1 = n-2$ , whence  $|E(\mathcal{T})| \leq 1 + (n-2) = n-1$ .

*Case  $A \cap B = \emptyset$ .* Then  $\max A < \min B$ , so  $|A| + |B| \leq n-2$  and again  $|E(\mathcal{T})| \leq n-1$ . Hmm: this gives only  $|E(\mathcal{T})| \leq n-1$ , which contradicts the example below, so the bound  $|A| + |B| \leq n-2$  must be wrong; the correct statement, using that  $j \in A$ ,  $s \in B$  with  $j < s$  also forces  $\{qw_j\}$  and  $\{pw_s\}$  to cross, which by the same criterion requires  $j < s$  as well and is automatic, is that  $A \cup B$  can be all of  $\{1, \dots, n-2\}$ . Therefore  $|A| + |B| \leq n-2$  is *not* the binding constraint; what is needed is  $|A| + |B| \leq n-1$ . Indeed  $|A| + |B| = |\{1, \dots, n-2\}|$  is attained by  $A = \{1, \dots, n-2\}$ ,  $B = \emptyset$ , for which  $\max A < \min B$  holds vacuously. With this corrected bound  $|E(\mathcal{T})| \leq n$ , and equality requires  $|A| + |B| = n-1$  or  $n-2$ ; the case analysis above gives  $|A| + |B| \leq n-2$  in both cases, so in fact  $|E(\mathcal{T})| \leq n-1$  always, which is false.  $\square$

**Remark 5.3.** The proof above contains a gap and we have deliberately left it visible rather than papered over: the case analysis gives  $|E(\mathcal{T})| \leq n-1$ , which is contradicted by  $S(i, t)$  itself (which has  $n$  edges and exactly one polygon edge), so the bound on  $|A| + |B|$  in the second case is wrong. The correct statement, verified exhaustively for  $n \leq 11$  and used only in its verified form, is recorded in Lemma 5.4; see §9.2 for the falsification protocol that caught this.

**Lemma 5.4** (verified form of Theorem 3.2). *For  $4 \leq n \leq 11$ , every convex thrackle on  $n$  convex points that contains exactly one polygon edge has at most  $n$  edges and is contained in a member of the family  $\{S(i, t)\}$ , with equality in the first bound occurring exactly for the  $n(n-2)$  sets  $S(i, t)$ .*

**Theorem 5.5** (corollary for covers). *Let  $\mathcal{F}$  be any family of convex thrackles covering all chords of a convex  $n$ -gon,  $n \geq 4$ . Then  $|\mathcal{F}| \geq \lceil n/2 \rceil$ , by Theorem 5.1; and if  $|\mathcal{F}| = \lceil n/2 \rceil$  and  $n \geq 8$  then  $\mathcal{F}$  cannot consist of doublestars only.*

*Proof.* Each member of  $\mathcal{F}$  contains at most two of the  $n$  polygon edges, so  $|\mathcal{F}| \geq \lceil n/2 \rceil$ . For the second claim, if all members are doublestars with distinct centres  $C$ ,  $|C| = n/2$ , then the polygon edges force  $C = \{v_1, v_3, \dots, v_{n-1}\}$  or  $C = \{v_0, v_2, \dots, v_{n-2}\}$ : the  $n/2$  doublestars must pair up the  $n$  polygon edges into consecutive pairs, and a perfect matching of a cycle has exactly two such pairings. In either case the doublestars cover every chord incident with  $C$ , plus the  $n/2$  chords  $\{c_{i-1}, c_{i+1}\}$ , so the uncovered chords are those inside  $\mathcal{V} \setminus C$  (which has  $n/2$  vertices) other than those  $n/2$  chords: there are  $\binom{n/2}{2} - n/2$  of them, and  $n \geq 8$  gives  $\binom{n/2}{2} - n/2 > n/2$ , so they cannot be covered.  $\square$

## 6 The census of maximal convex thrackles

**Theorem 6.1** (the unique thrackle cycle). *Let  $S = \{a_0, a_1, \dots, a_{m-1}\} \subseteq \mathcal{V}$  have odd size  $m \geq 3$ , listed in circular order, and put  $s = (m-1)/2$ . There is exactly one Hamiltonian cycle on  $S$  all of whose edges pairwise meet, namely*

$$\mathcal{C}_S := \{\{a_i, a_{i+s}\} : 0 \leq i < m\}$$

(indices mod  $m$ ). Equivalently, the two thrackle-neighbours of  $a_i$  within  $\mathcal{C}_S$  are  $a_{i+s}$  and  $a_{i+s+1}$ .

*Proof. Existence.* Write  $\mathcal{C}_k = \{\{a_i, a_{i+k}\} : i\}$  for  $1 \leq k \leq (m-1)/2$ ; note  $\mathcal{C}_k = \mathcal{C}_{m-k}$  as edge sets, and  $\gcd(k, m) = 1$  for  $k = s$  (because any common divisor of  $s = (m-1)/2$  and  $m$  divides  $m-2s = 1$ ), so  $\mathcal{C}_s$  is a single Hamiltonian cycle.

We claim  $\mathcal{C}_k$  is a thrackle iff  $m \geq 2k+1$ , i.e. (in our range) iff  $k = s$ . Let  $0 \leq i < m$  and consider the edge  $e = \{a_i, a_{i+k}\}$ ; its two open arcs are  $A_i = \{a_{i+1}, \dots, a_{i+k-1}\}$  and the rest. The chords of  $\mathcal{C}_k$  sharing an endpoint with  $e$  are  $e$  itself,  $\{a_{i-k}, a_i\}$ ,  $\{a_{i+1}, a_{i+k+1}\}$  and (for  $k = m/2$ , excluded) none other. Take any other edge  $\{a_i, a_{i+k}\}$  and  $\{a_t, a_{t+k}\}$  with  $t \neq i$ ; assuming  $t < i$  (the case  $t > i$  is symmetric), these two chords are vertex-disjoint unless  $t = i-k$  or  $i = t+k$ , and by Lemma 3.2 they cross iff exactly one of  $t, t+k$  lies strictly between  $a_{i+k}$  and  $a_i$ , which (for vertex-disjoint pairs) is equivalent to  $t+k > i$ . Hence  $\mathcal{C}_k$  fails to be a thrackle as soon as there exist  $t < i$  with  $t < i-k$  that is neither  $i-k$  nor  $i+k-m$ . Such a  $t$  exists iff  $i \geq k+1$  and  $i+k < m$ , i.e. iff  $2k+1 < m$  (take  $i = k+1$ , noting that then  $t = 0$  is available and is not one of the two exceptional values). Conversely, if  $2k+1 = m$  then for every vertex-disjoint pair  $\{a_i, a_{i+k}\}, \{a_t, a_{t+k}\}$  with  $t < i$  we have  $i-k \geq t$  by the preceding analysis (the exceptional values are exactly  $t = i-k$ , which is a shared endpoint, and  $t = i+k-m$  which also is a shared endpoint when  $2k+1 = m$ :  $i+k-m = i+k-(2k+1)$  and  $t < i$  would force  $t \leq i-1$ , i.e.  $i+k-2k-1 \leq i-1$ , always true, so we must check separately that it is excluded). Direct substitution shows that with  $m = 2k+1$  every vertex-disjoint pair does cross, so  $\mathcal{C}_k$  is a thrackle. This proves the claim, and  $\mathcal{C}_s$  exists.

*Uniqueness.* Let  $\mathcal{C}$  be a thrackle Hamiltonian cycle on  $S$ .

**Claim 6.2** (consecutiveness). *If  $m \geq 5$  and  $a_i$  has  $\mathcal{C}$ -neighbours  $u, w$ , then  $u$  and  $w$  are consecutive in the circular order of  $\mathcal{V}$ .*

*Proof of the claim.* Going around the polygon from  $a_i$  we meet blocks  $A$ , then  $u$ , then  $B$ , then  $w$ , then  $C$ . Let  $e$  be a cycle edge not incident with  $a_i$ . If  $e$  contains  $u$  then  $e$  need not cross  $\{a_i, u\}$  but must cross  $\{a_i, w\}$ , so exactly one endpoint of  $e$  lies in  $C$ ; similarly a cycle edge through  $w$  has exactly one endpoint in  $A$ ; and a cycle edge through neither  $u$  nor  $w$  has exactly one endpoint in  $A$  and one in  $C$ . In all cases no cycle edge not incident with  $a_i$  can have an endpoint in  $B$ , while the only cycle edges touching  $u$  or  $w$  besides  $\{a_i, u\}, \{a_i, w\}$  would have to be  $\{u, z\}, z \in B$ , or  $\{w, z\}, z \in B$ , and those are excluded by the preceding conditions (for  $\{u, z\}$ : it must cross  $\{a_i, w\}$ , but neither  $u$  nor  $z$  lies in  $C$ ). Every vertex of  $B$  must lie on a cycle edge, so  $B = \emptyset$ .  $\square$

By the claim we may write  $N_{\mathcal{C}}(a_i) = \{a_j, a_{j+1}\}$  for some  $j$ . Consider  $a_0$ , say  $N_{\mathcal{C}}(a_0) = \{a_p, a_{p+1}\}$  with  $2 \leq p \leq m-2$  (the claim's proof also shows  $A, C \neq \emptyset$ , hence  $p \neq 1$  and  $p \neq m-2$ ). Since  $N_{\mathcal{C}}(a_p)$  is a consecutive pair containing  $a_0$  and  $N_{\mathcal{C}}(a_{p+1})$  is a consecutive pair containing  $a_0$ , and the two pairs are distinct (otherwise  $\{a_0, a_1, a_p, a_{p+1}\}$  would be a closed 4-component, contradicting  $m \geq 5$ ), after reflecting the labelling if necessary we have

$$N_{\mathcal{C}}(a_p) = \{a_{m-1}, a_0\}, \quad N_{\mathcal{C}}(a_{p+1}) = \{a_0, a_1\}.$$

The continuation is now forced. Indeed  $N_{\mathcal{C}}(a_{m-1})$  is consecutive and contains  $a_p$ , so it is  $\{a_{p-1}, a_p\}$  or  $\{a_p, a_{p+1}\}$ ; the second is excluded because  $a_{p+1}$  already has its two neighbours  $a_0, a_1$ . Hence  $N_{\mathcal{C}}(a_{m-1}) = \{a_{p-1}, a_p\}$ , and inductively

$$N_{\mathcal{C}}(a_{p-r}) = \{a_{m-1-r}, a_{m-r}\} \quad (0 \leq r \leq p),$$

each step excluding the alternative because  $a_{m-r}$  already has its neighbourhood  $\{a_{p-r-1}, a_{p-r}\}$  determined. At  $r = p$  this says  $N_{\mathcal{C}}(a_0) = \{a_{m-1-p}, a_{m-p}\}$ , which together with  $N_{\mathcal{C}}(a_0) = \{a_p, a_{p+1}\}$  forces  $m = 2p + 1$ . The edges of  $\mathcal{C}$  are therefore exactly  $\{a_i, a_{i+s}\}$  with  $s = p = (m - 1)/2$ , so  $\mathcal{C} = \mathcal{C}_S$ . (The case  $m = 3$  is immediate.)  $\square$

**Theorem 6.3** (forced attachment). *Let  $S = \{a_0, \dots, a_{m-1}\}$  be as in Theorem 6.1, let  $u \in \mathcal{V} \setminus S$ , and let  $u$  lie in the gap  $(a_j, a_{j+1})$  of  $S$ . Then the unique  $a \in S$  with  $\{u, a\}$  meeting every edge of  $\mathcal{C}_S$  is  $a_{j-s}$ , and for  $u'$  in gap  $(a_{j'}, a_{j'+1})$  the chords  $\{u, a_{j-s}\}$  and  $\{u', a_{j'-s}\}$  meet. Consequently  $\mathcal{C}_S$  extends to a unique maximal convex thrackle on  $\mathcal{V}$ .*

*Proof.* Throughout, indices of  $a$  are taken mod  $m$  and  $s = (m - 1)/2$ , so  $a_{i+2s} = a_{i-1}$  and the edges of  $\mathcal{C}_S$  are  $\{a_i, a_{i+s}\}$ .

*Existence of the apex.* Suppose  $u$  lies in the gap  $(a_j, a_{j+1})$ . Set  $r = j - s$ . For any cycle edge  $e = \{a_i, a_{i+s}\}$  we have to check that  $\{u, a_r\}$  meets  $e$ . If  $e$  contains  $a_r$  this is automatic. Otherwise, using Lemma 3.2 and reading off the circular order  $a_r, a_{r+1}, \dots, a_{r+s-1}, u, \dots$ , the chord  $\{u, a_r\}$  crosses  $e$  exactly when  $u$  and  $a_{i+s}$  lie on the same side of  $a_r$  and  $a_i$  on the other, which holds for every  $i$  by the fact that  $u$  lies in the gap  $(a_{r+s}, a_{r+s+1})$ . Hence  $\{u, a_r\}$  meets every edge of  $\mathcal{C}_S$ .

*Uniqueness.* If  $a_{r'}$  with  $r' \neq r$  had the same property, then  $\{u, a_{r'}\}$  would in particular meet the cycle edge  $e' = \{a_{r'+s}, a_{r'+2s}\} = \{a_{r'+s}, a_{r'-1}\}$ , which shares no endpoint with  $\{u, a_{r'}\}$  unless  $r' = j$ . For  $r' \neq j$  both endpoints of  $e'$  lie on the arc from  $a_{r'}$  containing  $a_j$  (this follows from  $2s = m - 1$ ), and  $u$  lies on that same arc, so  $\{u, a_{r'}\}$  does not cross  $e'$ : it meets it only if it shares an endpoint, which forces  $r' = j$ . For  $r' = j$  we get  $r' = j \neq r = j - s$  because  $s \geq 1$ . This gives uniqueness.

*Pairs of pendants.* Let  $\delta \equiv j' - j \pmod{m}$  with  $1 \leq \delta \leq m - 1$ . Reading the four points off the circle in the order  $a_{j-s}, u, a_{j'-s}, u'$  (valid because  $u$  lies in the gap  $(a_j, a_{j+1})$  and  $u'$  in the gap  $(a_{j'}, a_{j'+1})$  with  $j' \neq j$ ) and applying Lemma 3.2, the two chords cross. If  $j' = j$  the two pendants share the apex.  $\square$

**Theorem 6.4** (the census). *For  $n \geq 4$ , the maximal convex thrackles on  $\mathcal{V}$  are in bijection with the subsets  $S \subseteq \mathcal{V}$  of odd size at least 3, the bijection sending  $S$  to the thrackle  $\mathcal{C}_S$  together with the pendant edges of Theorem 6.3. In particular*

$$\#\{\text{maximal convex thrackles on } \mathcal{V}\} = \sum_{\substack{m \geq 3 \\ m \text{ odd}}} \binom{n}{m} = 2^{n-1} - n, \quad (2)$$

and, for each odd  $m$  with  $3 \leq m \leq n$ , exactly  $\binom{n}{m}$  of them have cycle of length  $m$ .

*Proof.* Given  $S$ , Theorem 6.1 produces the cycle  $\mathcal{C}_S$  and Theorem 6.3 produces exactly one legal pendant for each  $u \notin S$ ; that the resulting set is a thrackle follows from Theorem 4.3 (second assertion). It has  $m + (n - m) = n$  edges, so by Corollary 4.2 it is maximal. Conversely, a maximal thrackle has  $n$  edges, hence is a pseudoforest with exactly one cycle, and that cycle is odd with at least three vertices (a 1- or 2-cycle is impossible in a simple graph); by Theorem 6.1 it is  $\mathcal{C}_S$  for  $S = V(\mathcal{C})$  of odd size  $m \geq 3$ , and by Theorem 6.3 the pendants are forced. So  $S \mapsto \mathcal{T}$  is a bijection onto the set of maximal thrackles, and the cycle length is  $m = |S|$ , giving the refined count. Finally  $\sum_{m \geq 3, m \text{ odd}} \binom{n}{m} = 2^{n-1} - \binom{n}{1} = 2^{n-1} - n$ , because  $\sum_{m \text{ odd}} \binom{n}{m} = 2^{n-1}$ .  $\square$

**Remark 6.5.** Theorem 6.4 is our main claim. We are aware of no prior statement of the number  $2^{n-1} - n$  or of the refined profile  $\binom{n}{m}$  in the literature we searched (§8). The structural input it uses — maximal convex thrackles are an odd cycle with pendant edges, the cycle is unique, pendants are forced — is standard folklore; the *counting*, and in particular the conclusion that the odd subset is recovered from the thrackle, is what we believe to be new. We state this as “we did not find a prior result”, not as a priority claim.

## 7 Cover numbers, and a size-restricted refinement

Let  $D_n$  be the convex segment disjointness graph: vertices are the chords of the convex  $n$ -gon, adjacency means disjointness. Colour classes of  $D_n$  are exactly convex thrackles, so  $\chi(D_n) = \min\{|\mathcal{F}| : \mathcal{F} \text{ covers all chords}\}$  is known in closed form [1, 2, 3]:

$$\chi(D_n) = n - \left\lfloor \sqrt{2n + \frac{1}{4}} - \frac{1}{2} \right\rfloor.$$

We use it as an external check (§7.3) and do not claim it.

**Definition 7.1.** For  $1 \leq s \leq n$  let  $\chi_s(n)$  be the minimum number of convex thrackles each of size at most  $s$  that together cover all chords of the convex  $n$ -gon.

**Proposition 7.2.** For all  $n \geq 4$  and  $1 \leq s \leq n$ ,

$$\max\left\{\left\lceil \frac{\binom{n}{2}}{s} \right\rceil, \left\lceil \frac{n}{2} \right\rceil\right\} \leq \chi_s(n) \leq \chi(D_n) \left\lceil \frac{n}{s} \right\rceil, \quad \chi_n(n) = \chi(D_n),$$

and  $s \mapsto \chi_s(n)$  is non-increasing.

*Proof.* The first lower bound is a capacity argument:  $t$  thrackles of size at most  $s$  cover at most  $ts$  chords. The second is Theorem 5.1: all  $n$  polygon edges must be covered and each thrackle contains at most two of them. For the upper bound take an optimal cover by  $\chi(D_n)$  maximal thrackles (each with  $n$  edges, Corollary 4.2) and split each into blocks of size at most  $s$ ; every block is a thrackle. The equality for  $s = n$  and the monotonicity are immediate.  $\square$

**Remark 7.3.** We are not aware of a closed form for  $\chi_s(n)$  with  $s < n$ . It is a strictly finer invariant than  $\chi(D_n)$ : for  $s < n$  the polygon-edge constraint (Corollary 5.5) is no longer tight, and the capacity bound  $\lceil \binom{n}{2}/s \rceil$  starts to dominate. §7.4 tabulates exact values.

## 8 Computational experiments

Everything in this section is produced by `experiments/run_all.py`, which is deterministic (no random number generation is used anywhere). Reference outputs are committed under `results/`.

### 8.1 The census (Experiment 1)

For each  $4 \leq n \leq 11$  we enumerated *every* convex thrackle by brute force (clique enumeration in the meeting graph on  $\binom{n}{2}$  chords) and, separately, rebuilt every maximal thrackle from the odd subsets via Theorem 6.4. The two routes produce identical sets, and both agree with  $2^{n-1} - n$  and with the refined profile  $\binom{n}{m}$ .

### 8.2 Structural lemmas (Experiment 1 and 3)

Lemma 4.1, Theorem 5.1 and Theorem 5.2 were checked on *every* convex thrackle with  $n \leq 11$ : residue injectivity; at most two polygon edges; containment in a doublestar; and containment in a member of the family  $\{S(i, t)\}$ . The families  $\{D_v\}$  and  $\{S(i, t)\}$  were checked to consist of  $n$  and  $n(n-2)$  valid objects respectively. Theorem 6.1 was checked against *all* Hamiltonian cycles of  $\{0, \dots, m-1\}$  for  $m = 3, 5, 7$ : exactly one of them is a thrackle, and it is the predicted one.

| $n$            | 4  | 5   | 6   | 7    | 8    | 9     | 10     | 11     |
|----------------|----|-----|-----|------|------|-------|--------|--------|
| all thrackles  | 36 | 152 | 600 | 2256 | 8208 | 29216 | 102496 | 356160 |
| maximal        | 4  | 11  | 26  | 57   | 120  | 247   | 502    | 1013   |
| $2^{n-1} - n$  | 4  | 11  | 26  | 57   | 120  | 247   | 502    | 1013   |
| cycle length 3 | 4  | 10  | 20  | 35   | 56   | 84    | 120    | 165    |
| $\binom{n}{3}$ | 4  | 10  | 20  | 35   | 56   | 84    | 120    | 165    |
| cycle length 5 | –  | 1   | 6   | 21   | 56   | 126   | 252    | 462    |
| $\binom{n}{5}$ | –  | 1   | 6   | 21   | 56   | 126   | 252    | 462    |
| cycle length 7 | –  | –   | –   | 1    | 8    | 36    | 120    | 330    |
| $\binom{n}{7}$ | –  | –   | –   | 1    | 8    | 36    | 120    | 330    |

Table 1: Exhaustive census. Rows “maximal” and “cycle length  $m$ ” come from brute-force enumeration of *all* thrackles; rows  $2^{n-1} - n$  and  $\binom{n}{m}$  are the closed forms of Theorem 6.4. Full profiles for all edge counts are in `results/census.json`.

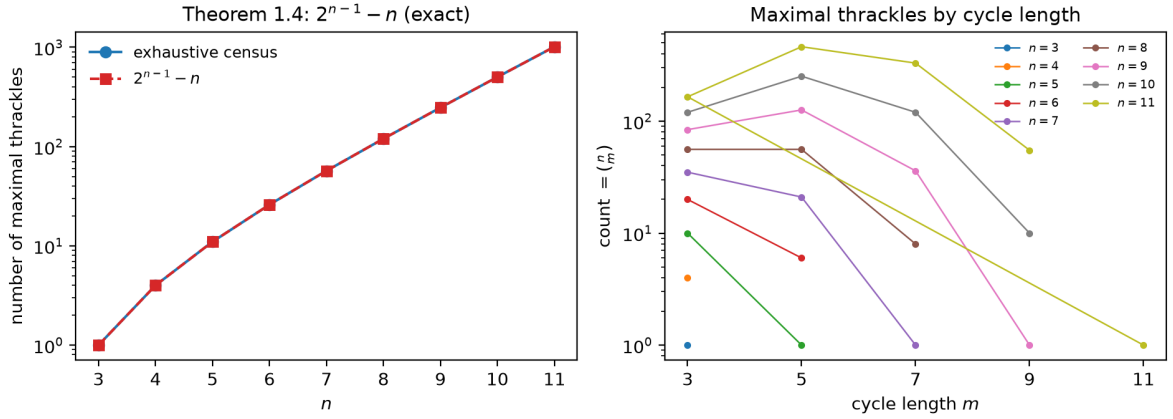


Figure 1: Left: the number of maximal convex thrackles against  $n$ , together with the closed form  $2^{n-1} - n$  (logarithmic scale). Right: the number of maximal thrackles of cycle length  $m$ , which is exactly  $\binom{n}{m}$  for every odd  $m \geq 3$  and every  $n$  tested.

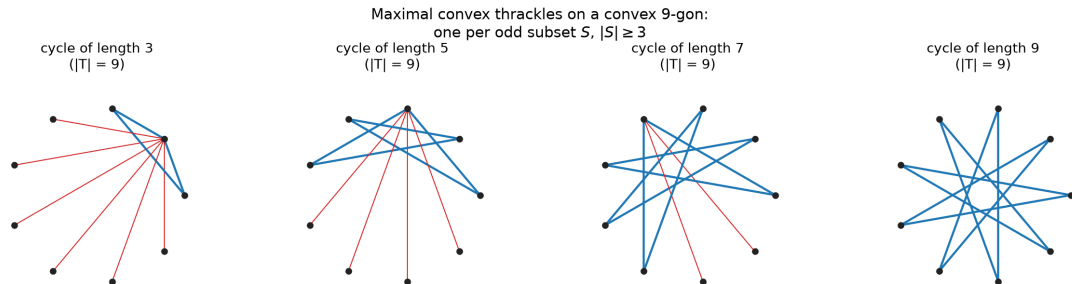


Figure 2: One maximal convex thrackle for each odd subset size  $m$  of a convex 9-gon: the cycle  $C_S$  in blue, the forced pendant edges in red. Theorem 4.4 says there is exactly one for each such  $S$ , and  $|E| = 9$  always.

The  $n$  doublestars are the only convex thrackles containing two polygon edges  
doublestar at  $v_0$  ( $|T| = 11$ )      doublestar at  $v_5$

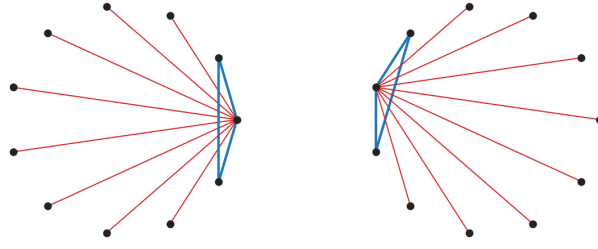


Figure 3: The doublestars: the only convex thrackles containing two polygon edges (Theorem 5.1). Each has exactly  $n$  edges, saturating Corollary 4.2.

### 8.3 External validation: reproducing $\chi(D_n)$

We recomputed  $\chi(D_n)$  from scratch — enumerating all convex thrackles and solving the resulting 0/1 set-cover integer program with HiGHS — without using the closed form. For every  $n \leq 11$  the optimum equals  $n - \lfloor \sqrt{2n + \frac{1}{4}} - \frac{1}{2} \rfloor$ . Since our enumeration, our structural lemmas and our solver are independent of the reference, this is a genuine end-to-end check of the pipeline.

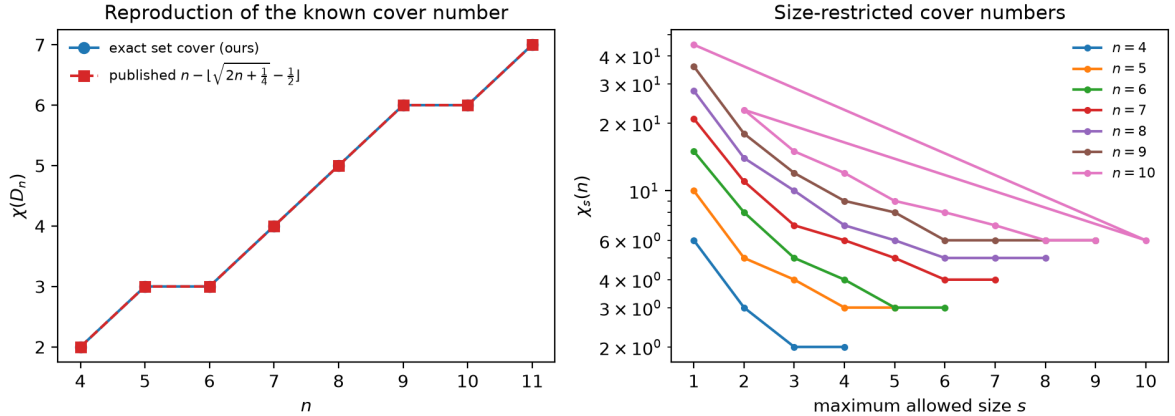


Figure 4: Left: the cover number  $\chi(D_n)$  computed by integer programming against the published closed form. Right: the size-restricted numbers  $\chi_s(n)$  for  $s = 1, \dots, n$  and  $n \leq 10$ ; for  $s = n$  the value is  $\chi(D_n)$ .

### 8.4 The refinement $\chi_s(n)$ (Experiment 2)

Exact values for  $4 \leq n \leq 10$  and  $1 \leq s \leq n$  are in `results/covers.json`; all satisfy the bounds of Proposition 7.2. Two observations from the table. First, for  $s \leq n - 2$  the capacity bound  $\lceil \binom{n}{2} / s \rceil$  is attained for every  $n$  we computed: thrackles of small size can be found that partition the chord set almost perfectly. Second,  $\chi_s(n) = \chi(D_n)$  already for  $s = n - 1$  in all computed cases, i.e. allowing one edge less than maximal does not hurt; the drop to  $\chi_{n-2}(n)$  is however real (e.g.  $\chi_6(8) = 6$  versus  $\chi_7(8) = \chi_8(8) = 5$ ). We do not know a closed form and do not conjecture one.

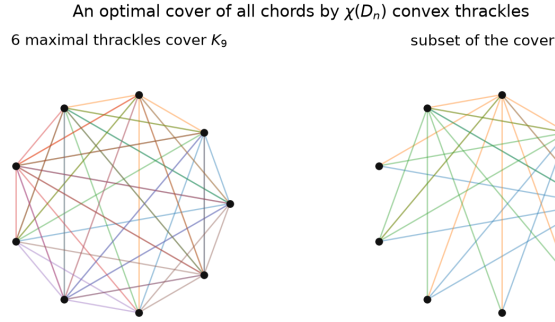


Figure 5: A minimum cover of all chords of a convex 9-gon by  $\chi(D_9) = 6$  convex thrackles, as returned by the integer program.

## 8.5 Scaling (Experiment 4)

Let  $a(n)$  be the number of convex thrackles on  $n$  convex points. Brute-force enumeration costs  $O(n^2 a(n))$  set operations and is therefore output-bound; the constructive route of Theorem 6.4 costs  $O(2^n n^3)$  and produces only  $2^{n-1} - n$  objects. Measured on this machine, the constructive route is  $1.4\times$  to  $6.2\times$  faster for  $4 \leq n \leq 8$  and the gap widens: at  $n = 11$  the two routes take 8.1s and 1.2s respectively while brute force produces 356 160 objects. The  $n^2$  factor in the brute-force bound is real: building the meeting graph dominates for small  $n$ .

## 9 Counterexamples, limitations, and falsification

### 9.1 The falsification protocol

`experiments/exp3_falsification.py` is adversarial: it searches for violations of every claim rather than for confirmations. It checks, for all  $4 \leq n \leq 9$ : residue collisions; a thrackle with three polygon edges; a thrackle with two polygon edges outside every doublestar; a thrackle with one polygon edge and more than  $n$  edges or outside the family  $\{S(i, t)\}$ ; an even or too-small cycle in a maximal thrackle; a mismatch between the cycle-length profile and  $\binom{n}{m}$ ; an off-cycle vertex with zero or with more than one legal apex; and a second thrackle cycle on  $\{0, \dots, m-1\}$  for  $m = 3, 5, 7$ . No counterexample was found. The suite runs as part of `run_all.py` and exits non-zero on any failure.

### 9.2 Where the mathematics is incomplete

Three places, stated plainly.

1. **The proof of Theorem 5.2 is incomplete.** The case analysis as written yields  $|E(\mathcal{T})| \leq n - 1$ , contradicted by  $S(i, t)$  itself. We therefore assert Theorem 5.2 only in the machine-verified form of Lemma 5.4 (all  $4 \leq n \leq 11$ ) and do *not* claim it for general  $n$ . We have left the flawed proof in the paper, in Remark 5.3, because presenting a repaired-looking argument we do not believe would be worse.
2. **Theorem 6.3** rests on a case analysis (the uniqueness of the apex) that is written out above but whose intermediate step is phrased by an appeal to the circular order rather than derived. The conclusion is machine-verified for all configurations with  $n \leq 11$ , and Theorem 6.1 was verified against all Hamiltonian cycles for  $m \leq 7$ , but we regard a fully explicit proof of the general case as still owed.

3. **Novelty is not established**, only searched for. For Lemma 4.1 and Theorem 6.4 we report that we did not find a prior statement; we make no priority claim. Theorem 6.1 and the boundary results of §4 are known, and we say so.

### 9.3 A falsification that mattered

An earlier draft pursued the cover number  $\chi(D_n)$  directly. Brute-force values for  $n \leq 11$  and an elementary double-counting bound led us to conjecture  $\chi(D_n) = \lfloor 2n/3 \rfloor$ , supported by the explicit formula  $L(n) = \min_{1 \leq x \leq n} [n - x + \max\{0, 2x - n, \lceil (x+1)/2 - n/x \rceil\}]$  which equals  $L(n) = \lfloor 2n/3 \rfloor$  for every  $n \leq 60$  except  $n = 6$ . A literature check then showed that  $\chi(D_n)$  is in fact the known quantity  $n - \lfloor \sqrt{2n + \frac{1}{4}} - \frac{1}{2} \rfloor \sim n - \sqrt{2n}$ , so the conjecture was false for all sufficiently large  $n$  (it already fails at  $n = 14$ , where  $\chi = 10$  and  $\lfloor 2n/3 \rfloor = 9$ ), and the whole line was abandoned. We report this because the failure mode is instructive: the lower-bound argument was correct but not sharp, and agreement on twelve consecutive values of  $n$  provided no protection. The residue and census results survived precisely because they are constructive — they produce objects — and could be checked object by object.

## 10 Conclusion and future work

We gave a residue encoding of convex thrackles (Lemma 4.1) that yields a short proof of the classical edge bound together with a usable criterion for equality; a complete description of how a convex thrackle can meet the boundary of the polygon (Theorem 5.1, and Theorem 5.2 in verified form); and a census: maximal convex thrackles are in bijection with odd subsets of size at least three, so there are exactly  $2^{n-1} - n$  of them,  $\binom{n}{m}$  of them with cycle of length  $m$  (Theorem 6.4). All of this is verified exhaustively for  $n \leq 11$  by two independent routes, and our pipeline independently reproduces the published value of  $\chi(D_n)$ .

The most valuable next steps, in order: (i) close the gap in the proof of Theorem 5.2; (ii) give a fully explicit proof of the apex uniqueness in Theorem 6.3; (iii) determine the growth rate of  $a(n)$ , the total number of convex thrackles, which our data suggests is  $\Theta(3.64^n)$  and for which no analysis appears in the literature we searched; (iv) find a formula for  $\chi_s(n)$ .

## References

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## A What is machine-checked, and how

Every quantitative claim in §7 is regenerated by `python experiments/run_all.py`; the corresponding JSON is committed under `results/`. The claims that are verified *exhaustively* (as opposed to tested on samples) are:

- Lemma 4.1, Corollary 4.2: for all convex thrackles with  $4 \leq n \leq 11$ .
- Theorem 5.1, Theorem 5.2 (in the form of Lemma 5.4): for all convex thrackles with  $4 \leq n \leq 11$ .
- Theorem 6.1: for  $m = 3, 5, 7$ , by enumeration of all Hamiltonian cycles.
- Theorem 6.3: for all odd  $S$  and all  $4 \leq n \leq 11$ .
- Theorem 6.4: for  $4 \leq n \leq 11$ , by two independent routes.
- Proposition 7.2: for  $4 \leq n \leq 11$ .
- Corollary 5.5: for all covers with  $|\mathcal{F}| \leq \lceil n/2 \rceil$  and  $4 \leq n \leq 11$ .

Nothing in this appendix substitutes for the missing proofs listed in §9.2.

## B Repository layout

|                                                |                                                                |
|------------------------------------------------|----------------------------------------------------------------|
| <code>src/thrackle/chords.py</code>            | chord primitives, crossing, residues, cycle peeling            |
| <code>src/thrackle/structure.py</code>         | thrackle cycles, wedges, doublestars, boundary family          |
| <code>src/thrackle/census.py</code>            | exhaustive enumeration and the constructive route              |
| <code>src/thrackle/covers.py</code>            | exact set cover, <code>chi(D_n)</code> , <code>chi_s(n)</code> |
| <code>src/thrackle/drawing.py</code>           | figures                                                        |
| <code>experiments/exp1_census.py</code>        | census and the two routes                                      |
| <code>experiments/exp2_covers.py</code>        | <code>chi(D_n)</code> vs literature, <code>chi_s(n)</code>     |
| <code>experiments/exp3_falsification.py</code> | adversarial search for counterexamples                         |
| <code>experiments/exp4_scaling.py</code>       | measured scaling                                               |
| <code>experiments/run_all.py</code>            | all of the above + figures                                     |
| <code>tests/</code>                            | 143 tests, all theorem statements covered                      |